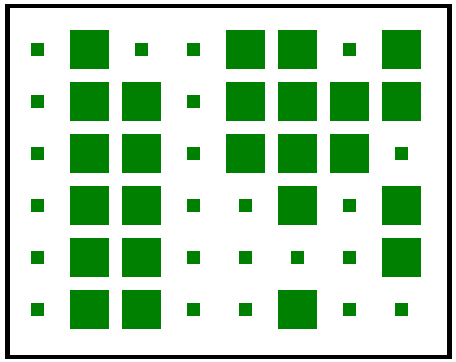
Note: Students who previously took CS230 may not also receive credit for CS330.



Class 1
Course Overview & Introduction

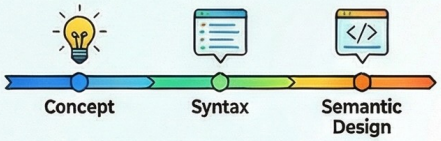
CS 330
Programming Languages and Compilers

CS330: Your Guide to Compilers & Language Design

Skidmore College: A comprehensive exploration of programming languages, from design principles to compiler implementation, blending core theory with modern AI integration and collaborative projects.

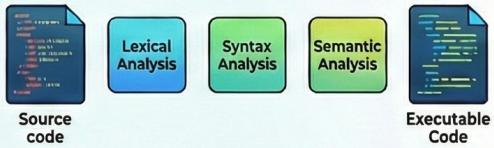
WHAT YOU'LL LEARN

Master the Full Lifecycle of Programming Languages



Go from concept to application by exploring language categories, syntax, and semantic design.

Implement a Compiler From Scratch

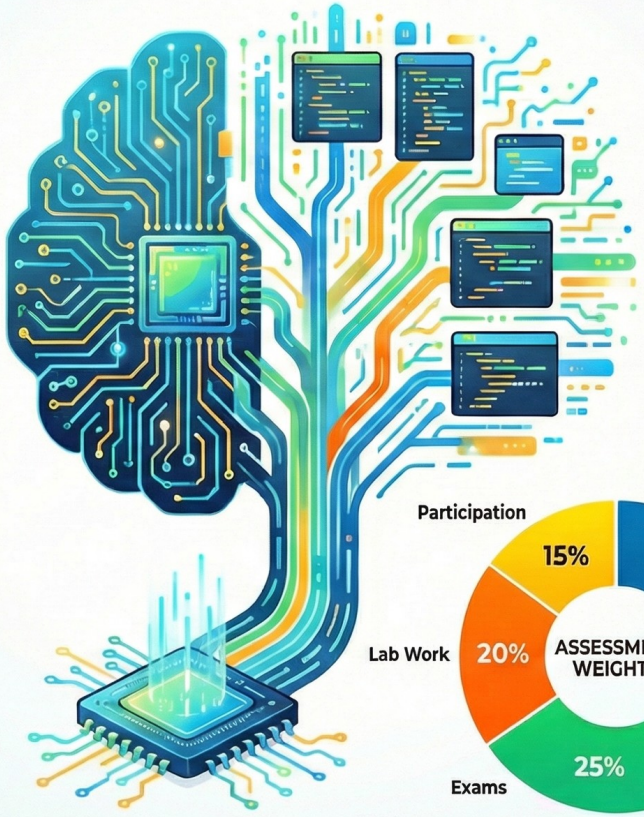


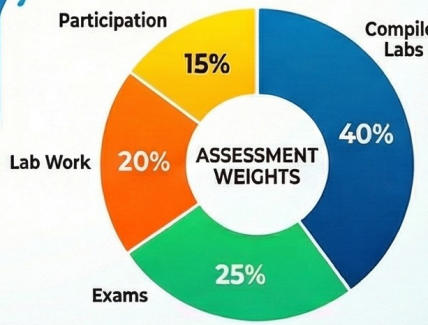
Convert source code to executable code through lexical, syntax, and semantic analysis.

Code Across Multiple Paradigms



Learn the key differences between imperative, object-oriented, functional, and declarative languages.





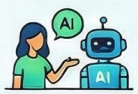
HOW YOU'LL LEARN & SUCCEED

Integrate an AI Language Model into Your Workflow

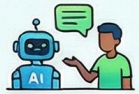
A course-provided tool lets you leverage an LLM for design, development, and testing.



I've got this!
You may not use the LLM in any way for the assignment.



Vibe coding!
You may leverage and modify LLM-generated output to complete the lab.



You do it!
The LLM must generate all artifacts; you only refine prompts.

Collaborate in Structured, Semester-Long Teams



Mirror industry practice with rotating roles like Chief Understanding and Chief Equity Leader.

Benefit from Student-Focused Policies



2 x 3-DAY EXTENSION COUPONS



OPTIONAL IN-CLASS MINDFULNESS RESETS

Use two 3-day extension coupons and participate In optional in-class mindfulness resets.

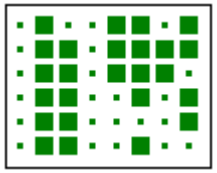
NotebookLM

Instructor Background

- Music Education
- Computer Science: BS/MS
- Learning/Education: EdD
- GE – 13 years
- Blue Slate (consulting) – 15 years
- Skidmore – 13 years
- Enterprise systems
  - Finance, manufacturing, health care
  - Application Dev., Architecture, and Data (Security, Analytics, Semantics)
- CISSP, CBDE, GSEC, MongoDB, SCJP, RHCE



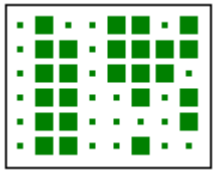
My Style



- Teaching Assumptions
  - Students and I want to learn
- Teaching Objectives
  - Intuition
  - Find useful documentation/standards
  - Give a person a fish, feed them for a day…
- Approach
  - Overview
  - Hands-on
  - Simplify to core -> add complexity

Syllabus Review

Note: Students who previously took CS230 may not also receive credit for CS330.



The Programming Languages and Compilers Syllabus

Saratoga Springs, NY | Fall, 2026 | *Your best source for initial CS330 insight*

Course Goal Announced; Students Anxious to Learn

"Expand students' expertise in the lifecycle of computer programming languages. These are human-made abstractions for communicating between people and people as well as people and computers. Exploring language categories, syntax, semantic design, and compiler implementation, takes students from concept to application," was Dave Read's response when queried about the point of the course. "Language designers differ in their goals which influence a language's fit for particular purposes. With these insights, students will become adept at learning new programming languages, creating languages for specific goals, and choosing appropriate languages for future projects, ultimately expanding a student's ability to create performant, easily-maintained, and reliable software."

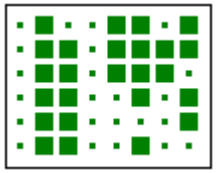
Learning Objectives Provide Course Focus

Dave provided additional detail by listing several objectives for the course. We've listed them here so that students may look to them when seeking clarity regarding the class.

Students will…

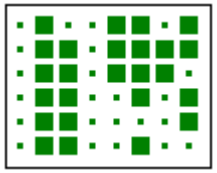
- Learn the major issues and choices language designers and implementers face when creating a programming language, including evaluation of language features based on relevant criteria
- Discern the key differences between imperative, object-oriented, functional, and declarative programming languages, writing programs in each
- Effectively leverage AI as a programming tool
- Clearly distinguish between syntax and semantics
- Implement a compiler, converting source code to executable code, exploring key steps such as lexical, syntax, and semantic analysis, as well as

Key Points



- Some lectures will be recorded and assigned as homework
- Some class time will be used for roundtable discussions and lab work
- I am free to call on you to answer questions and offer insights regarding any reading or video assignment
  - Responses inform your participation grade
- The use of an LLM does not replace a requirement that you understand and can apply course content, it augments it, adding another programming language family: LLMs

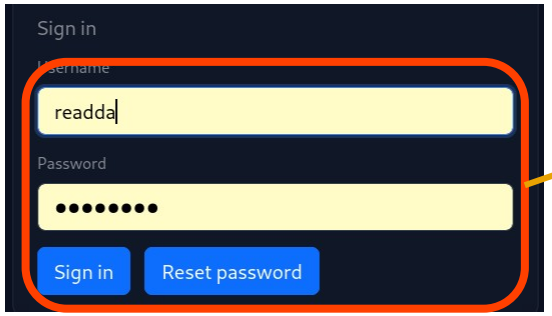
LLM Frontend Tool (BitBuddy Application)



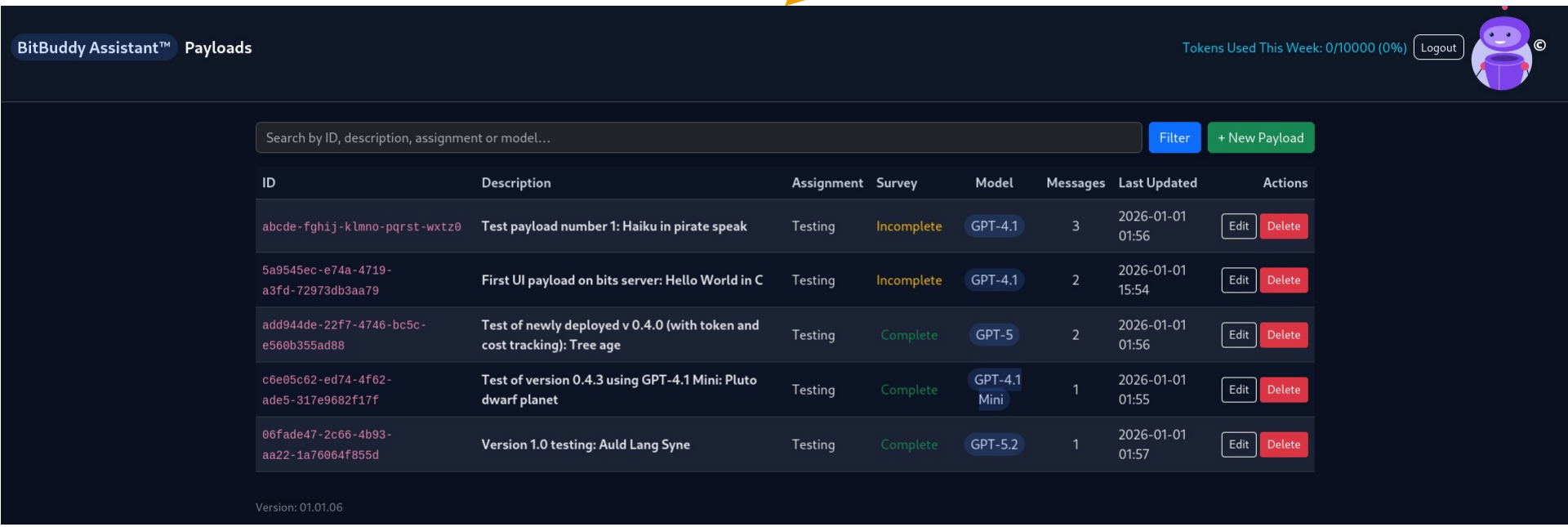
- Thank the Spring 2026 Programming Languages class for testing this app for us
  - Report bugs and enhancement requests in bug tracking (TBD)
- Do not take any action in the tab while the LLM connection is open, indicated by animated icon in center of screen, otherwise response will be lost
- o3 licenses have not been purchased
- All use of an LLM for lab work MUST use this front end so that we have an accurate data set of LLM interactions germane to our class

BitBuddy: Login

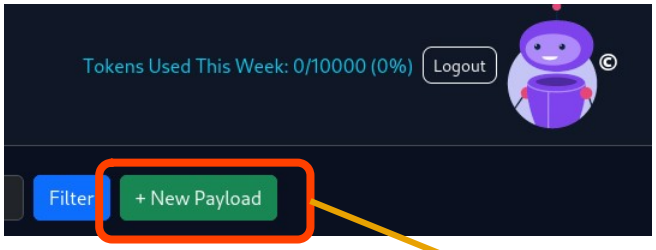
http://141.222.36.194/llm-frontend/

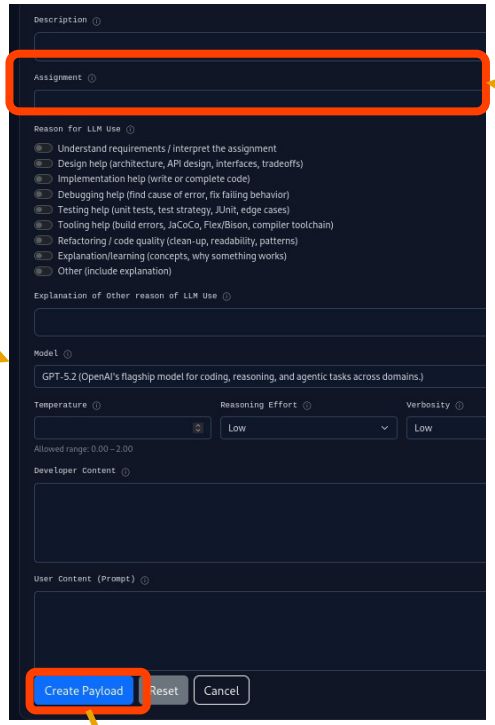


Existing payloads are listed, if applicable

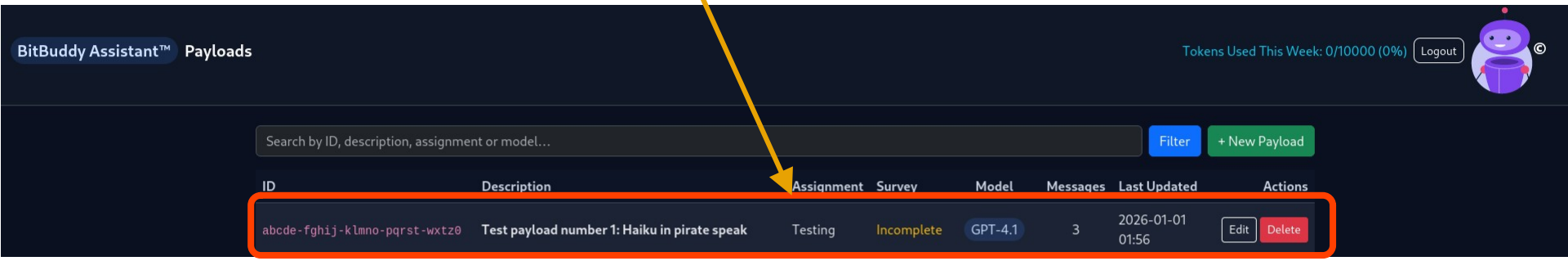


BitBuddy: Create Payload (No History)

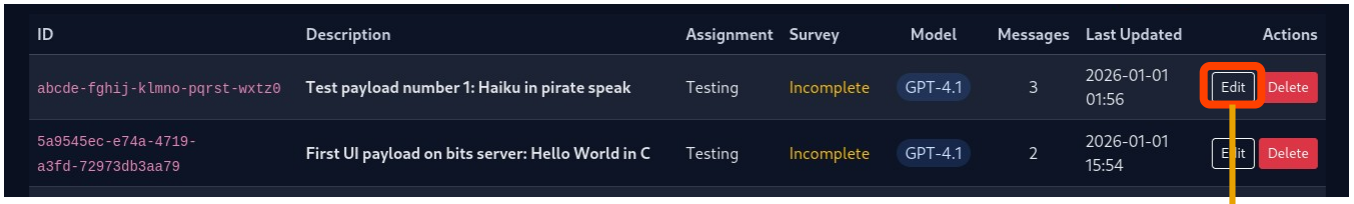




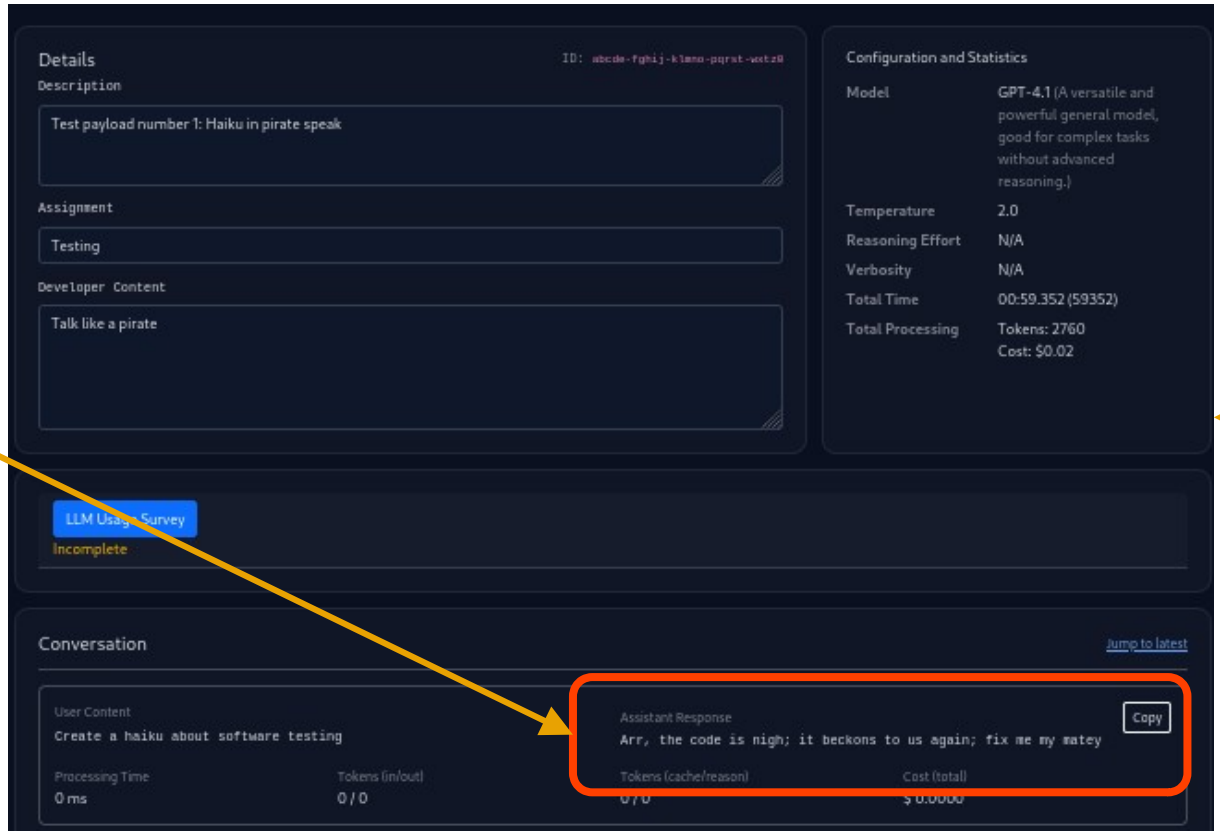
Include **Lab** and **Exercise** (or **Step**) identifiers in the "Assignment" input field. For example
Lab 2, Exercise 1



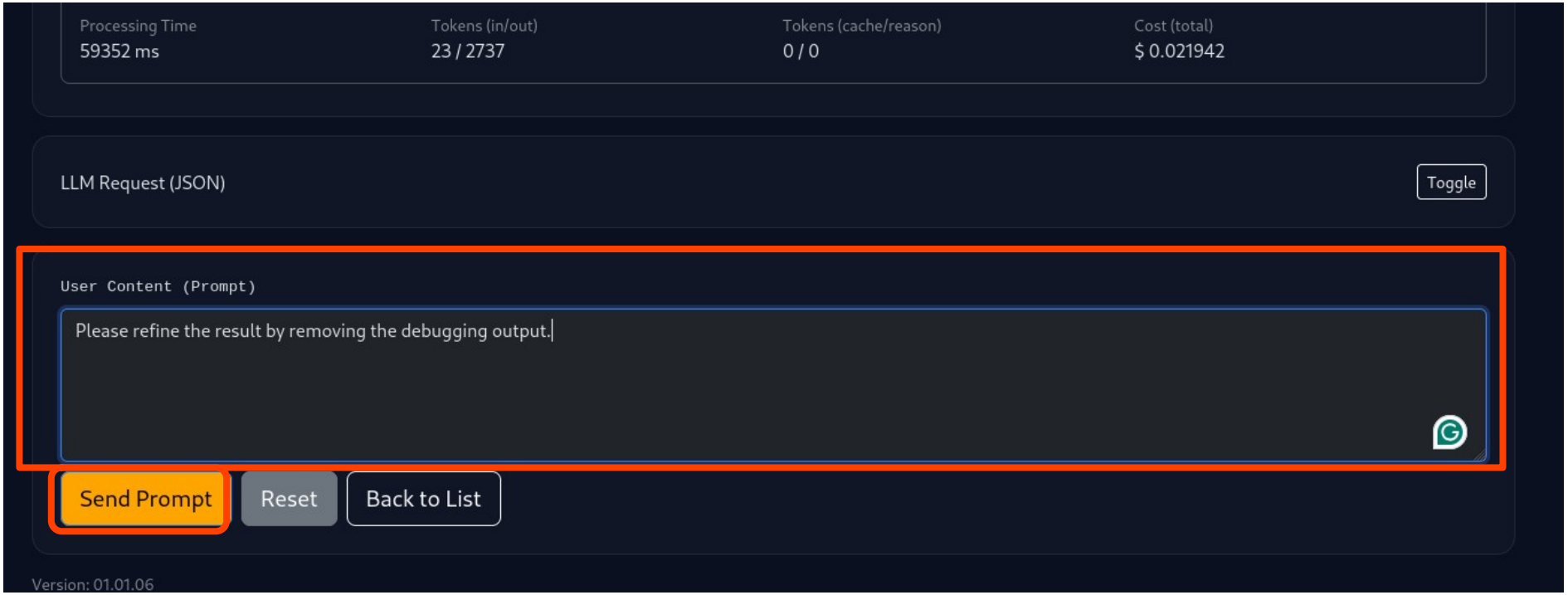
BitBuddy: View Response/Edit Payload



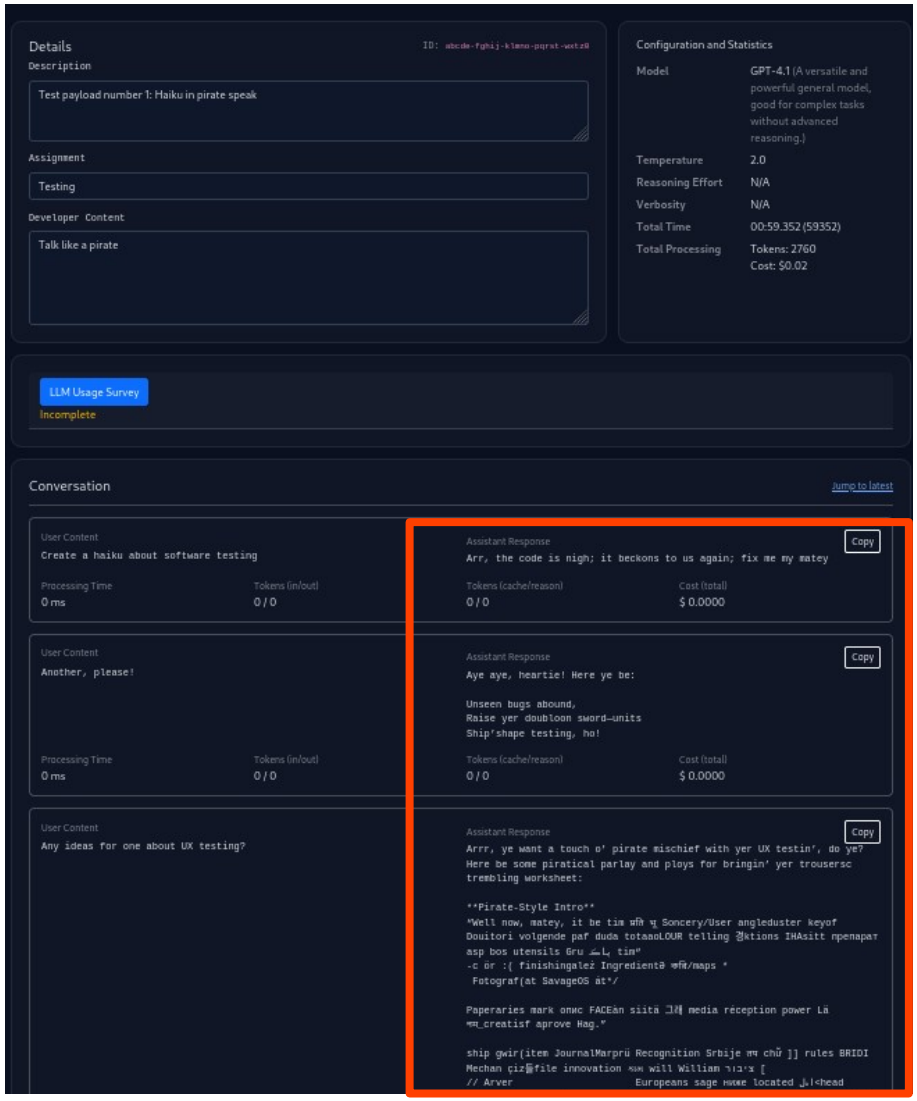
Assistant Response



BitBuddy: Continue Conversation

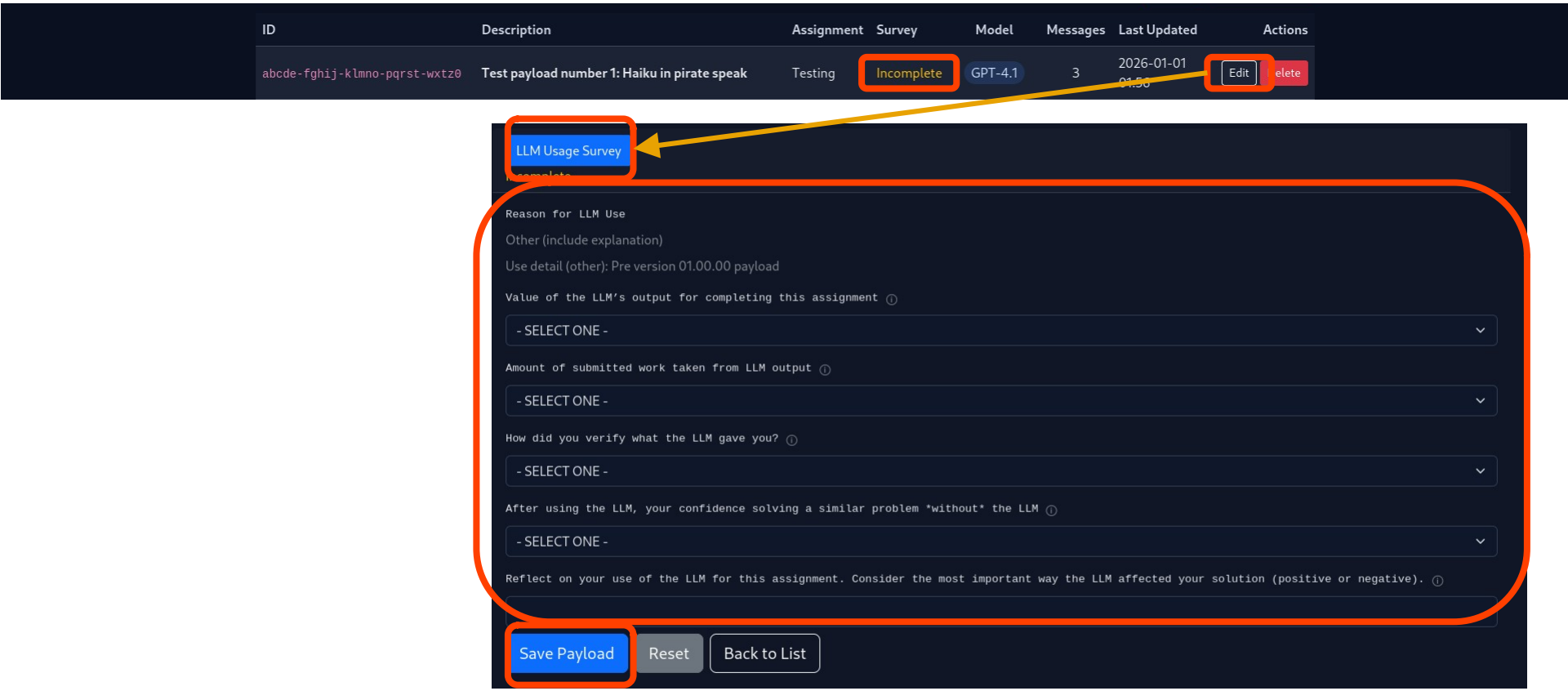


BitBuddy: Multiple Interactions

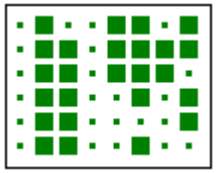


BitBuddy: Usage Survey

- Once you are done working with the payload, you must complete the LLM Usage Survey. Must complete survey before assignment will be accepted



Clarifying the LLM/Lab Modes of Use



- **“I’ve got this!”** – No use of LLM for exercise
- **“Vibe coding!”** – Use LLM as team member for the assignment. **May modify LLM-generated output** to fulfill exercise requirements. Attribution must indicate the LLM versus human-created content.
- **“You do it!”** - The LLM must generate all exercise artifacts. The individual or team **may not modify the generated output** other than by refining the LLM prompt.
- Syllabus clarifies requiresments for using these modes for labs. **Lab README.md file must note the mode used.**

Textbook



It is free to download

May be handy to have locally for looking up information

Remember to annotate in Perusall for assigned readings

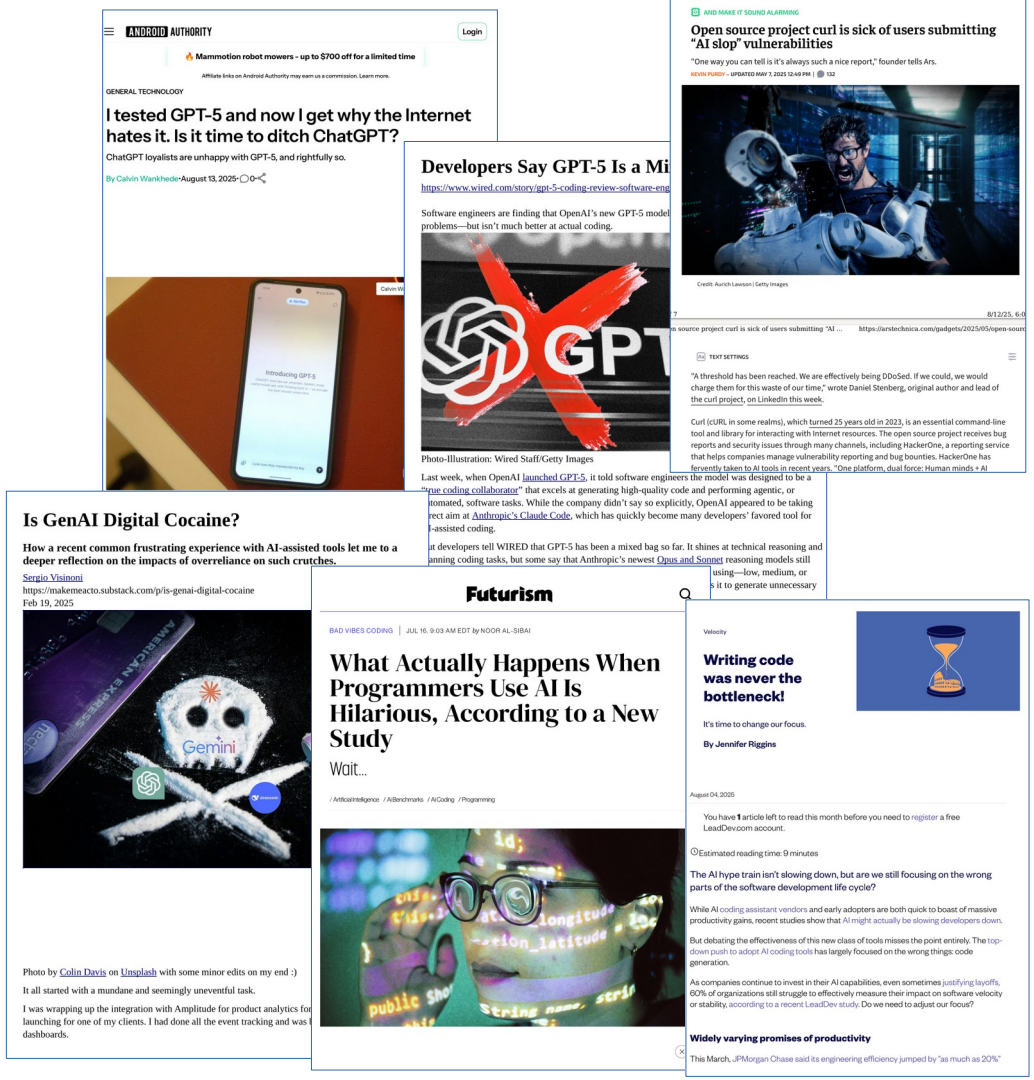
https://www3.nd.edu/~dthain/compilerbook/compilerbook.pdf

Additional Resources: LLMs

Each week, typically on Friday, we will discuss an article or video related to LLMs. Links are found in theSpring, and are due prior to class

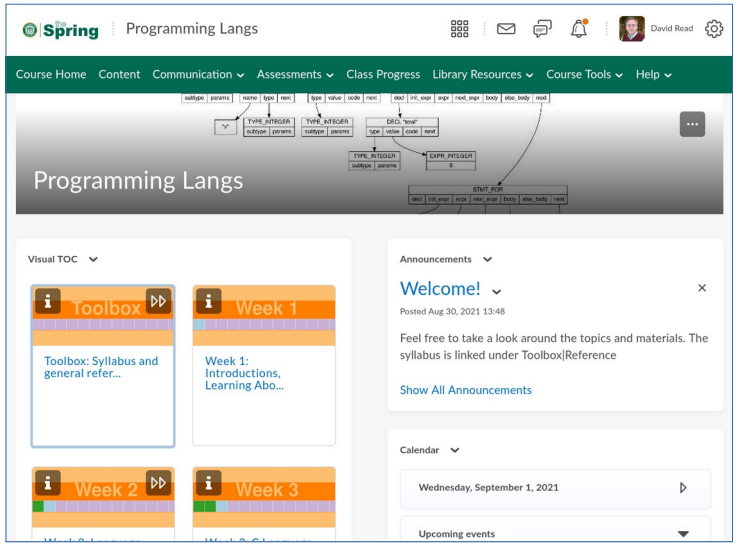
Perusall annotations are required

Be prepared to share your thoughts on the content in class



Glance at our Class Page on theSpring and Review Perusall Access

Tools



theSpring

Learning Management System

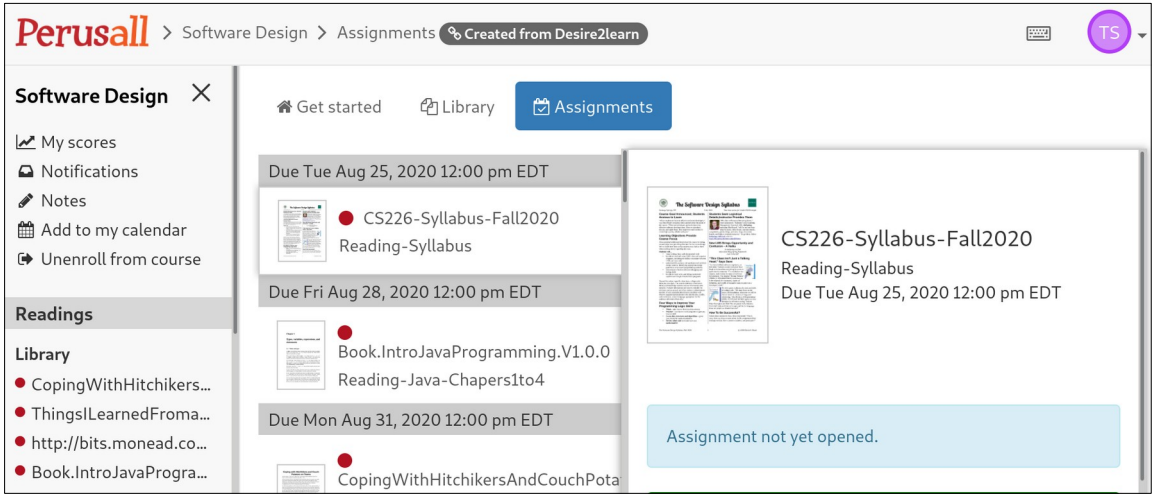
https://theSpring.skidmore.edu/

Perusall

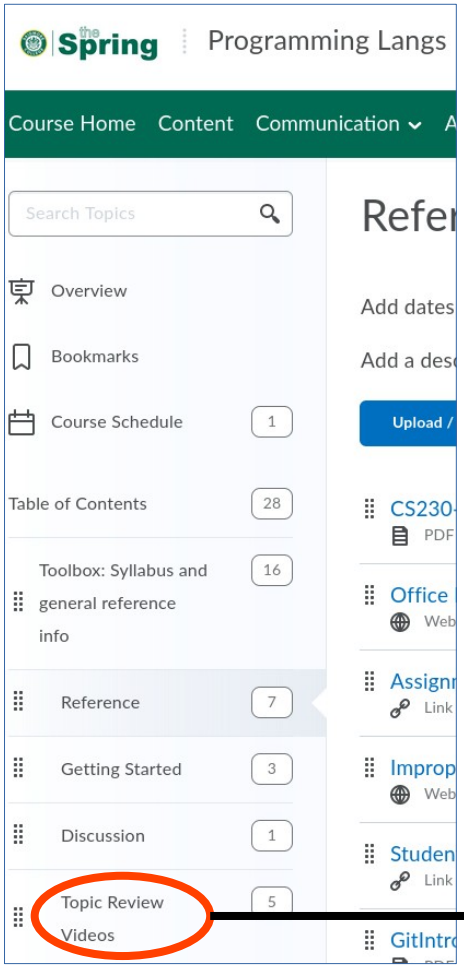
Collaborative Reading and Notetaking Platform

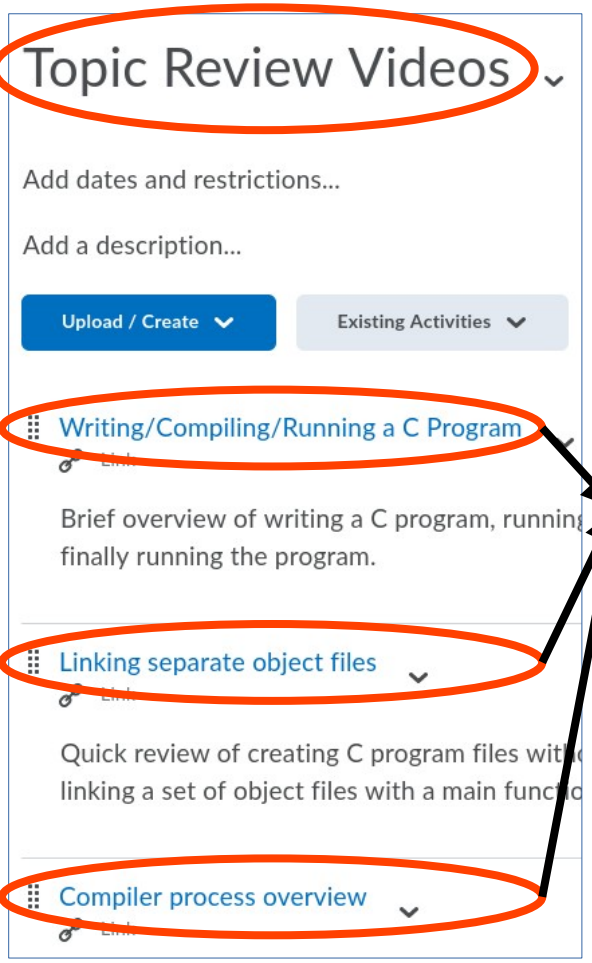
Linked in theSpring

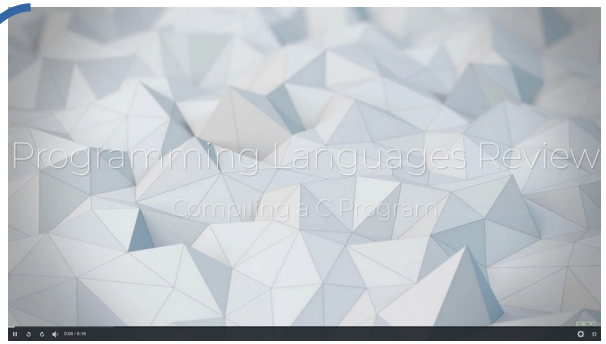
https://app.perusall.com/

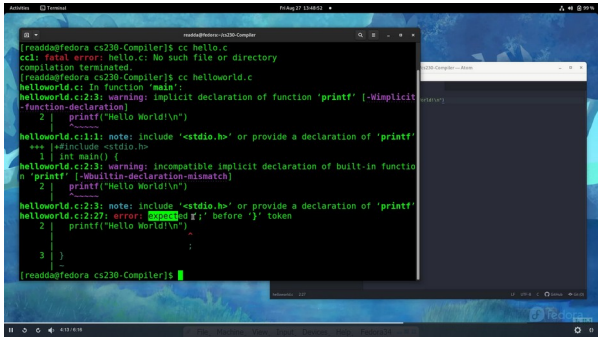


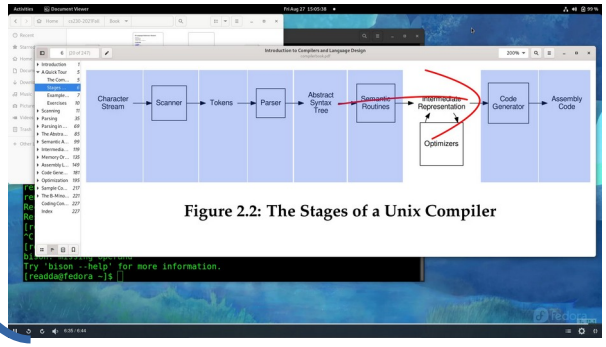
Review Videos











Teams

Housekeeping

- About You Survey



About You

Student Name

What is your preferred name?

What are your preferred pronouns
(e.g. she, her, hers; he, him, his; they, their, theirs...)

Is there anything you want to share regarding your background for this class (other computer science classes, self study...)

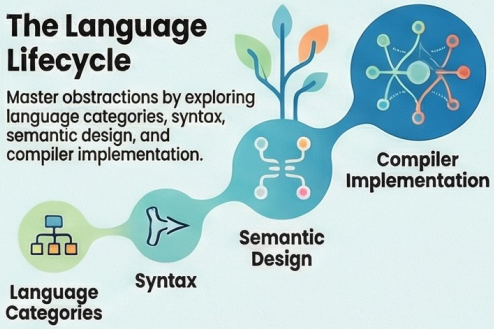
Are there any concerns you have about taking this class that you'd like to share?

CS 330: Programming Languages – Course Essentials

COURSE VISION & PHILOSOPHY

The Language Lifecycle

Master obstractions by exploring language categories, syntax, semantic design, and compiler implementation.



Intuition over Rote Learning

The "Teach a person to fish" approach prioritizes finding documentation and building standards.





Expert-Led Instruction

Led by Dr. Dave Read, featuring 30+ years of industry experience and historical computing expertise.

THE LLM FRAMEWORK & BITBUDDY

Use the BitBuddy Gateway



All LLM interactions for lobs MUST use the BitBuddy frontend for data tracking.

Three Modes of AI Use



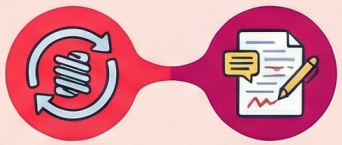


Mandatory Usage Surveys

You must complete on LLM Usage Survey before ony assignment will be accepted.

TOOLS & IMMEDIATE TASKS

Collaborative Learning Tools



theSpring

Utilize "theSpring" for course management.

Perusall

Utilize "Perusall" for collaborative reading and annotations.

The 2-1 Journal Rule

Weekly journol entries require a minimum of 2 paragraphs and 1 quiz question.



Week 1 Deadlines



- Complete the "About You" form.
- Week 1 Journal.
- Perusall annotations for Chapters 1-2.

NotebookLM

Homework

- Friday (9/11):
  - Syllabus (Perusall annotations)
  - Couch potatoes article and questionnaire (Perusall annotations)
  - LLM: Vibe Coding… Braindead Coders (Perusall annotations)
  - “About You” form
- Sunday (9/13):
  - Journal entry for week 1
    - **2-1 Rule** (minimum 2 paragraphs, 1 quiz question)
- Monday (9/14):
  - Compilers book Chapters 1 and 2
    - Perusall annotations!

